PCS Solution of the Hahn–Banach Extension Theorem

A concrete application of the Projection-Coordination System in functional analysis

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I Problem Statement

Hahn–Banach Extension Theorem.设X be a real linear space, $p : X \to \mathbb{R}$ a sublinear functional ($p(x+y) \le p(x) + p(y)$,$p(\alpha x) = \alpha p(x)$ for $\alpha \ge 0$). Let $Y \subseteq X$ be a subspace and $f : Y \to \mathbb{R}$ a linear functional satisfying $f(y) \le p(y)$对所有$y \in Y$. Then there exists a linear extension $F : X \to \mathbb{R}$ such that

$$F|_Y = f, \qquad F(x) \le p(x) \quad (\forall x \in X).$$

A foundational theorem in functional analysis, underpinning duality theory and weak topologies. The classical proof relies on Zorn's lemma and is therefore non-constructive.

II Classical Proof (Zorn's Lemma)

1. Partially ordered set.

$$\mathcal{E} = \{(Z, g) : Y \subseteq Z \subseteq X,\ g|_Y = f,\ g(z) \le p(z)\},$$

with $(Z, g) \preceq (Z', g')$ whenever $Z \subseteq Z'$以及$g'|_Z = g$.

2. Every chain has an upper bound. For a totally ordered subset $\{(Z_\alpha, g_\alpha)\}$, set $Z_\infty = \bigcup_\alpha Z_\alpha$以及$g_\infty(z) = g_\alpha(z)$ for $z \in Z_\alpha$. Well-definedness follows from the chain condition.

3. A maximal element is the whole space. If a maximal element $(Z_\star, g_\star)$ satisfied $Z_\star \ne X$, pick $x_0 \in X \setminus Z_\star$ and construct a proper extension to $\mathrm{span}\{Z_\star, x_0\}$, contradicting maximality.

Issue: Entirely non-constructive. Zorn's lemma is equivalent to the axiom of choice and provides no algorithm or approximation.

III PCS Encoding

Encode the extension problem as a Projection-Coordination System Σ_{HB}; existence is equivalent to the existence of a fixed point.

Component	Definition	Intuition
$\mathcal{I}$	$\{1\}$	Single local constraint bundle
X_1	X^* (algebraic dual)	Space of all linear functionals
M_1	$\{F \in X^* : F\|_Y = f\}$ (affine subspace)	Functionals satisfying the local condition
P_1	Orthogonal projection onto M_1	Correction to meet the local constraint
M_{glob}	$\{F : F(x) \le p(x)\ \forall x\}$ (p-ball)	Functionals satisfying the global bound
G	Projection onto M_{glob}	Compression into the globally feasible set
$\mathcal{T}_\Sigma$	$G \circ P_1$	Composition of local correction and global compression

Ω-obstruction tower. By the PCS uniformization theorem (Thm 9.1), $\mathrm{Fix}(\Sigma) \neq \emptyset \iff \{\omega_k\} = 0$. For Σ_{HB},$|\mathcal{I}| = 1$ implies Čech cohomology $H^{\ge 1} = 0$, hence the obstruction tower vanishes identically:

$$\omega_k(\Sigma_{\text{HB}}) = 0\ (\forall k \ge 1) \implies \exists F \in \mathrm{Fix}(\Sigma_{\text{HB}}).$$

In other words: compatibility of local and global constraints guarantees a non-empty intersection — the Hahn–Banach theorem in PCS formulation.

IV Concrete Example: $\mathbb{R}^3$

Setup. $X = \mathbb{R}^3$ (Euclidean norm), $Y = \{(x, y, 0)\}$ (xy-plane), $f(x, y, 0) = 2x - y$, $\|f\| = \sqrt{5}$; $p(x) = \sqrt{5}\,\|x\|$ (norm-preserving extension).

By the Riesz representation theorem, each F corresponds to $w \in \mathbb{R}^3$:$F(v) = w \cdot v$, $\|F\| = \|w\|$.

- $M_1 = \{w : w_x = 2,\ w_y = -1,\ w_z \in \mathbb{R}\}$ (affine line)
- $P_1(w) = (2, -1, w_z)$ (orthogonal projection onto M_1)
- $M_{\text{glob}} = \{w : \|w\| \le \sqrt{5}\}$ (closed ball)
- $G(w) = w \cdot \min(1, \sqrt{5}/\|w\|)$ (projection onto the ball)
- $\mathcal{T}(w) = G(P_1(w))$

Unique fixed point: $w^\star = (2, -1, 0)$,$F(x, y, z) = 2x - y$,$\|w^\star\| = \sqrt{5}$.

The PCS iteration $w^{(k+1)} = \mathcal{T}(w^{(k)})$ converges to this fixed point from any initial value. The damped scheme $w^{(k+1)} = (1 - \lambda) w^{(k)} + \lambda \mathcal{T}(w^{(k)})$ accelerates convergence.

V PCS vs. Classical Methods

The classical approach relies on Zorn's lemma (equivalent to the axiom of choice), guaranteeing existence by a maximality argument on a partially ordered set, but offering no constructive procedure. The PCS method encodes the extension problem as a projection-coordination system, decides existence via the Ω-obstruction tower, and provides an iterative construction.

Aspect	Classical Method (Zorn)	PCS Method
Foundational tool	Axiom of choice (non-constructive)	Ω-obstruction tower (constructive)
Existence criterion	Maximal element argument	Vanishing of obstruction classes $\omega_k = 0$
Constructivity	No explicit extension produced	Iteration converges to solution$G \circ P_1$
Unity	Each problem argued independently	Unified framework: constraintsPCS-towersolvability$\to\to\Omega\to$

VI Conclusion and Significance

The PCS reformulation of the Hahn–Banach extension theorem reveals a deeper unifying principle:

Compatibility of local and global constraints vanishing -obstruction tower

$$\Longleftrightarrow \Omega$$

This is more than a restatement of the same mathematical fact: it provides a **unified obstruction criterion** for constraint-extension problems. Traditionally, each extension problem (Hahn–Banach, Tietze, Carathéodory, Lax–Milgram, etc.) requires an independent existence argument; in the PCS language, they all share the same -tower criterion — the differences lie only in the encoding and the topological structure.Ω

Further developments: -obstruction arithmetic (Conjecture C1) • PCS spectral algebra (Conjecture C2) • Complexity dichotomy (Conjecture C6)Ω